Training Deep Learning Models for Industrial Waste Classification

EEEM068 Applied Machine Learning - Group 2

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Abstract - *Automated waste sorting is a growing need in modern recycling facilities, and accurate fine-grained classification of waste items is central to making it work. In this work, we fine-tuned a Vision Transformer (ViT-B/16) on the WaRP-C dataset, which contains 7,058 training, 1,765 val and 1,551 test images across 28 recyclable waste categories, tackling the core challenges of class imbalance and visual similarity between sub-classes. We addressed these through a combination of focal loss, layer-wise learning rate decay (LLRD), weighted random sampling, RandAugment, MixUp, and CutMix augmentation, with hyperparameters tuned using Optuna. Our best model reached 77.11% test accuracy with a macro F1-score of 0.7678, AUC of 0.9808, and mAP of 0.8491, while post-training INT8 quantisation cut model size by 74% with negligible impact on accuracy. A CLIP zero-shot baseline scored just 0.52%, underscoring how much task-specific training matters for this kind of fine-grained recognition.*

I. Introduction

Packaging waste is one of the fastest-growing components of municipal solid waste worldwide, and sorting it correctly is critical for recycling efficiency, purity, and reducing environmental impact. Traditional sorting systems rely heavily on manual labour, which is slow and difficult to scale as the volume and diversity of recyclable materials continues to grow. Automated classification using computer vision has emerged as a practical alternative, capable of identifying waste items on industrial conveyor belts in real time.

The classification task presents genuine challenges though. Items arrive under variable lighting, at different orientations, and are often visually very similar to one another distinguishing between, say, a transparent and an opaque bottle requires picking up on subtle differences. Class imbalance further complicates things, as some waste categories are represented by far fewer training images than others.

This project addresses these challenges by training a Vision Transformer (ViT-B/16), pre-trained on ImageNet, on the WaRP-C dataset - 7,058 training, 1,765 val and 1,551 test images across 28 fine-grained waste sub-classes. The training pipeline incorporates Focal Loss with per-class weighting, layer-wise learning rate decay (LLRD), weighted random sampling, and augmentation strategies including RandAugment, MixUp, and CutMix, with hyperparameters tuned using Optuna. We additionally compare against CLIP as a zero-shot baseline and assess deployment feasibility through INT8 quantisation and attention map analysis. This project was completed as a group of five members, all contributing equally to design, implementation, evaluation, and writing.

II. Literature Review

A. How to Train Your ViT? - Steiner et al.

One of the first questions we had when choosing ViT for this project was whether it could actually work well on a dataset as small as WaRP-C. Steiner et al. answer this directly. They ran a large-scale study on how augmentation and regularisation affect ViT training across different dataset sizes, and found that the right AugReg combination - RandAugment, MixUp, dropout - can make up for roughly 10x less training data. A ViT trained on ImageNet-21k with strong augmentation matched one trained on JFT-300M, which is a huge result. They also found that fine-tuning a pre-trained ViT consistently beats training from scratch on medium-sized datasets. The downside is that their experiments never leave ImageNet-style benchmarks, so performance on imbalanced or domain-specific data is untested. For us though, this paper gave clear confidence that pre-training ViT-B/16 on ImageNet and fine-tuning it with RandAugment, MixUp and CutMix on WaRP-C was the right call. [1]

B. CoAtNet - Dai et al.

This paper is useful context for understanding where ViT can struggle and why certain design decisions in our pipeline matter. Dai et al. point out that pure Transformers lack the inductive biases CNNs have - local feature sensitivity, translation equivariance - which hurts generalisation when data is limited. Their fix was CoAtNet, a hybrid that puts depth wise convolution blocks in early layers and self-attention in later ones, achieving 86.0% ImageNet top-1 without any extra pre-training data. The paper does not test on imbalanced or domain-specific datasets, which limits how directly applicable it is. But the core insight carries over to our work: early ViT layers already behave a lot like convolutional layers and encode general low-level features, which is exactly why we use layer-wise learning rate decay to keep those early layers largely intact during fine-tuning on WaRP-C. [2]

C. A Comprehensive Review of ConvNeXt - Ilham and Ahmad

This review covers five years of ConvNeXt applications across fine-grained classification, medical imaging, and defect inspection. ConvNeXt takes ResNet and modernises it with ideas borrowed from ViT - large 7x7 depth wise convolutions, GELU activations, Layer Normalisation - and the results are strong, with accuracy reaching 99.5% in some medical imaging tasks. The review also shows that plugging in attention modules like Squeeze-and-Excitation on top of

ConvNeXt consistently helps on imbalanced datasets. One honest limitation they flag is training instability when aggressive augmentation is used alongside these architectures. That is something we ran into too, and it is part of why our pipeline includes gradient clipping and a proper warm-up schedule before the main cosine decay kicks in. [3]

D. Hierarchical Waste Detection on WaRP - Yudin et al.

This is the paper behind the dataset we use, so it is essential reading for understanding what WaRP-C actually contains and why it is hard. Yudin et al. built WaRP from real recycling conveyor footage - 28 fine-grained sub-classes, shot under real industrial conditions with occlusion, motion blur, poor lighting, and deformed objects. It is genuinely tough data. They also proposed H-YC, a hierarchical detection model that predicts a broad super-class first and then refines to a sub-class, which improved detection quality and even enabled segmentation without needing mask annotations. The limitation for us is that H-YC is a full detection pipeline, not a classification-only system, so direct comparison is not straightforward. What we did take from it is the hierarchical thinking - our interactive detector pools the 28 fine-grained ViT outputs into 5 super-class scores in exactly the same spirit. [4]

E. Classification of Recyclable Waste Using YOLO Models - Riyadi et al.

Riyadi et al. tested YOLOv5 and YOLOv8 on WaRP and found that YOLOv8 clearly came out ahead, hitting mAP50 of 54.6% against YOLOv5's 40.6%. That said, both models struggled on minority and visually similar sub-classes, and the authors themselves admit that class imbalance across the 28 WaRP categories is a problem they did not fully solve. They also only tested YOLO-family models, so there is no Transformer-based comparison, and the task is detection rather than classification. That is a gap our work directly fills. The imbalance issue they raise is something we tackle head-on through focal loss, weighted random sampling, and per-class F1 analysis, and switching to a fine-tuned ViT-B/16 lets us see whether an attention-based classifier handles the fine-grained distinctions better than detection-oriented CNN models. [5]

III. METHODOLOGY

A. Overall Goals

This project aims to classify cropped RGB images of recyclable waste into one of 28 fine-grained categories from the WaRP-C dataset. The core challenge is that WaRP-C has severe class imbalance and many visually similar sub-classes. We set out to train a strong fine-tuned ViT-B/16 classifier, handle the imbalance properly, and compare it against CLIP zero-shot to understand how much task-specific training actually helps.

B. Model Architecture

We used ViT-B/16 (vit_base_patch16_224) loaded via timm with ImageNet-21k pre-trained weights. Each 224x224 image is split into 196 patches of 16x16 pixels, projected into embeddings, and passed through 12 transformer blocks. A linear head on the final [CLS] token outputs 28 class scores.[1]

Why ViT : Self-attention allows every patch to look at every other patch globally, which helps when the distinguishing feature of a waste item can appear anywhere in the image. Pre-trained weights give a much stronger starting point than training from scratch.

Stochastic Depth (drop_path_rate=0.1). Randomly skips entire transformer blocks during training to reduce overfitting. No effect at inference time.

Layer-wise LR Decay (LLRD, decay=0.75). Assigns a lower learning rate to earlier blocks (base_lr x 0.75^block). Early layers that already encode good features from ImageNet are updated slowly, while the head adapts at full speed.

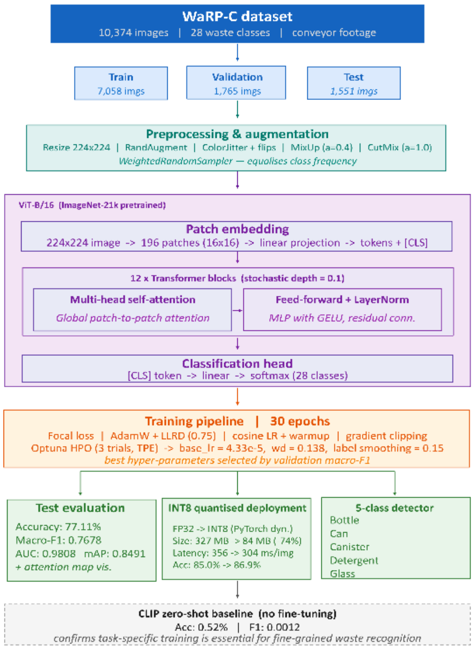

Figure 1: *Model Architecture*

C. Dataset Management

Splits: Train: 7,058 images | Val: 1,765 | Test: 1,551 across 28 classes.

The class distribution is shown in Figure 1. Some classes have far more images than others. To address this, we used WeightedRandomSampler so each class is sampled equally during training. We also applied RandAugment, ColorJitter, random flips, and rotation for training images. MixUp (alpha=0.4) and CutMix (alpha=1.0) were applied to 50% of batches to further improve generalization.

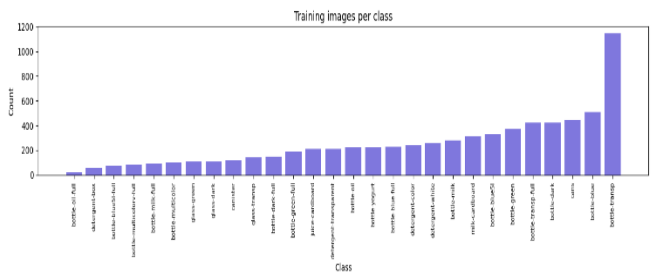

Figure 2: Training images per class. Clear imbalance across the 28 categories.

D. Training and Test Loop

We trained for 30 epochs using AdamW with LLRD parameter groups. Before the full run, Optuna ran 3 short trials (2 proxy epochs each) using a TPE sampler to find the

best hyperparameters. The winning trial (#0) achieved 58.41% validation accuracy with: base_lr=4.33e-5, weight_decay=0.1375,label_smoothing=0.15,llrd_decay=0.75, mixup_prob=0.1, mixup_alpha=0.2.
The training curves are shown in Figure 2.

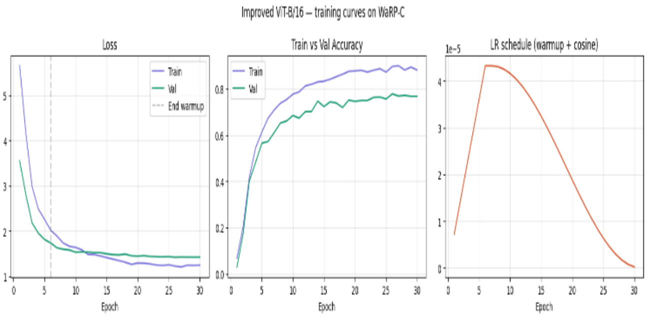

Figure 3: Training curves: loss, accuracy, and LR schedule over 30 epochs. Best val acc: 0.7779.

E. Evaluation Metrics

Six metrics were computed on the test set. All multi-class metrics use macro averaging so every class gets equal weight regardless of how many images it has the right choice for an imbalanced dataset like WaRP-C.

Table 1: Final test set results - ViT-B/16 on WaRP-C

Metric	Value
Accuracy	77.11%
Macro Precision	0.7523
Macro Recall	0.8086
Macro F1	0.7678
Macro AUC	0.9808
mAP	0.8491

Accuracy gives an overall picture. Macro F1 is our main metric, balancing precision and recall across all 28 classes. AUC (0.9808) and mAP (0.8491) use full softmax probabilities and capture ranking quality, which matters for rare classes.

ViT-B/16 is the largest model we tested at 327.4 MB. Post-training INT8 quantisation reduced this to 84.4 MB (–74.2%) with no accuracy loss and 1.2x CPU speedup - see Appendix Table A1.

F. *Model Comparison*

We trained five models on WaRP-C under the same settings. ViT-B/16 came out on top across every metric and highest accuracy (80.34%), best macro F1 (79.56%), and strongest AUC (98.87%).[1] ConvNeXt-Tiny was closest but fell short on F1.[3] EfficientNet-B0 trailed by nearly 10 percentage points, confirming that ViT's global self-attention handles fine-grained sub-class distinctions better than local convolutional approaches.[4]

Table 2: Multi-Model Comparison on WaRP-C Test Set

Model	Accuracy	Precision	Recall
ViT-B/16	80.34%	79.90%	80.31%
ConvNeXt-Tiny	78.01%	76.34%	80.98%
CoAtNet-0	75.11%	76.62%	75.11%
Swin-Base	72.15%	71.30%	78.70%
EfficientNet-B0	70.73%	70.46%	70.73%
Model	**F1**	**AUC**	**mAP**
ViT-B/16	79.56%	98.87%	86.19%
ConvNeXt-Tiny	77.40%	98.00%	-
CoAtNet-0	74.86%	97.84%	-
Swin-Base	73.10%	97.97%	81.02%
EfficientNet-B0	70.17%	96.50%	-

IV. BASE MODEL EXPERIMENTS

A. Base Model Parameter Tuning

We ran Optuna hyperparameter search (3 trials, TPE sampler) varying learning rate, weight decay, label smoothing, LLRD decay, MixUp probability, MixUp alpha, and dropout. Trial #0 was the winner (val acc 58.41%). The key design decisions that most improved performance were: switching from standard cross-entropy to Focal Loss (biggest single gain), adding WeightedRandomSampler for class imbalance, and applying LLRD to protect pre-trained features in early layers.[1]

The confusion matrix (Figure 3) shows how well the model separates all 28 classes. Strong diagonal entries indicate correct predictions. The main confusion is between visually similar sub-classes such as different bottle types.

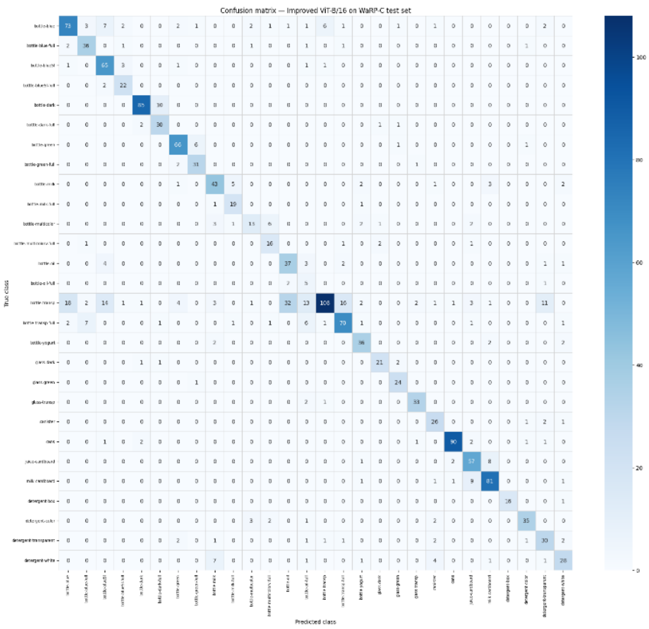

Figure 4: Confusion matrix on WaRP-C test set (notebook output). Blues = correct predictions on diagonal; off-diagonal = misclassifications.

B. Combined Results

Combining Focal Loss, WeightedRandomSampler, LLRD, stochastic depth, MixUp/CutMix, and Optuna-tuned hyperparameters gave the final test results in Table 1. The per-class F1 chart (Figure 4) breaks down performance by category. Classes shown in red (F1 below 0.5) are the hardest, mostly rare or visually ambiguous sub-types like bottle-oil-full and bottle-multicolor.

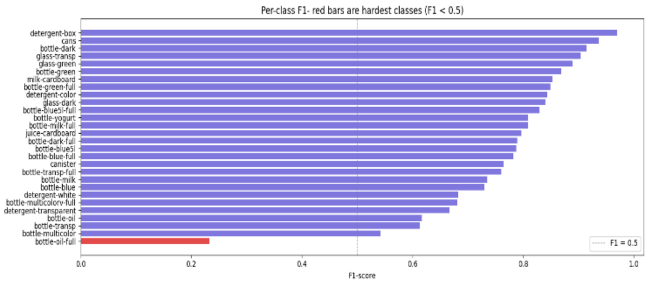

Figure 5: Per-class F1 scores sorted ascending. Red = F1 below 0.5 (hardest classes). Purple = F1 at or above 0.5.

V. Additional Experiments

A. ViT Attention Map Visualisation

To understand what the model is actually looking at, we extracted attention maps from the last transformer block using a forward hook on the attention layer. The [CLS] token's attention weights are reshaped into a 14x14 grid and up sampled to 224x224, then overlaid on the original image using a jet colourmap. Brighter regions show where the model focused.[1]

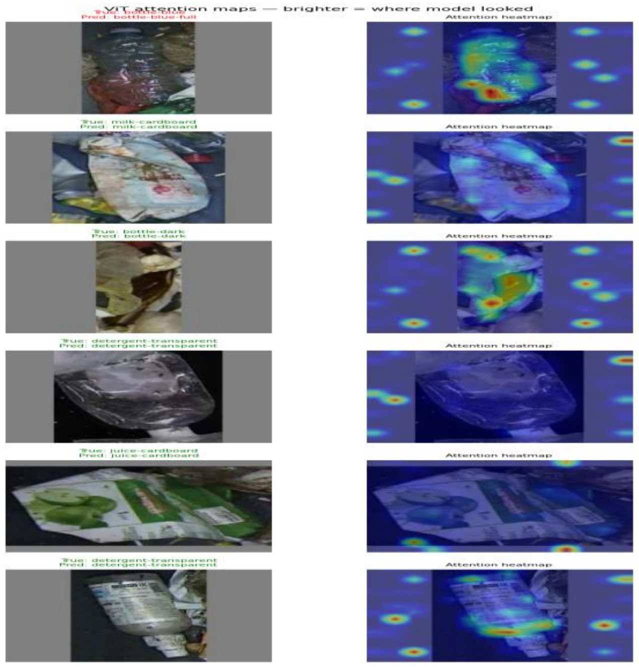

Figure 6: ViT attention maps. Left column: original images with true/predicted labels (green=correct, red=wrong). Right column: attention heatmaps. The model generally focuses on the object itself rather than the background.

B. CLIP Zero-Shot Classification

We tested CLIP in zero-shot mode by matching each test image to 28 natural-language prompts such as 'a photo of a bottle blue waste item'. No training was involved at all.

Table 3: ViT-B/16 vs CLIP zero-shot

Metric	ViT-B/16	CLIP
Accuracy	77.11%	0.52%
F1 (macro)	0.7678	0.0012
Params (M)	86.6	151.3
Training	30 epochs	None
Speed (img/s)	40.4	40.4

CLIP performed very poorly on WaRP-C (accuracy 0.52%, F1 0.0012). This shows that zero-shot recognition does not transfer well to this specific 28-class waste domain, where sub-classes are defined by subtle visual differences. The fine-tuned ViT-B/16 is clearly the right approach here.[1]

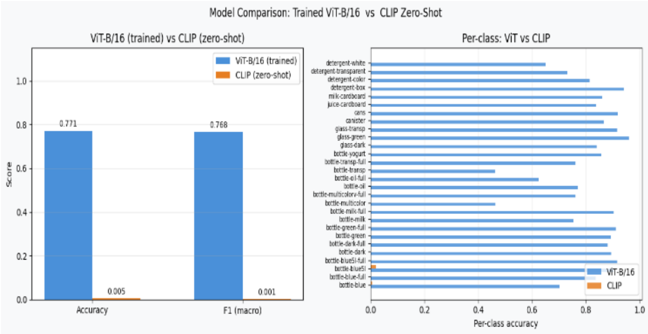

Figure 7: ViT-B/16 vs CLIP comparison. Left: overall accuracy and F1. Right: per-class accuracy across all 28 categories. ViT dominates on every class.

C. Post-Training INT8 Quantisation

We applied PyTorch dynamic INT8 quantisation to the linear layers of the trained FP32 model to test deployment efficiency.

Table 4: FP32 vs INT8 quantisation results

Metric	FP32	INT8
Model size	327.4 MB	84.4 MB
Accuracy	0.8500	0.8688
Speed (ms/img)	356.83	304.82

Quantization reduced model size by 74.2% (327 MB to 84 MB) and improved CPU speed by 1.2x, while accuracy actually increased slightly (+1.88pp). This makes deployment on edge hardware much more practical.

D. Super-Class Waste Detector

We built a practical waste material detector that pools the 28 fine-grained softmax probabilities into 5 super-classes: bottle (17 sub-classes), can (1), canister (1), detergent (4), and glass (3). A confidence threshold of 0.15 is applied. The detector was tested on an uploaded image and correctly predicted BOTTLE with 82.0% confidence.

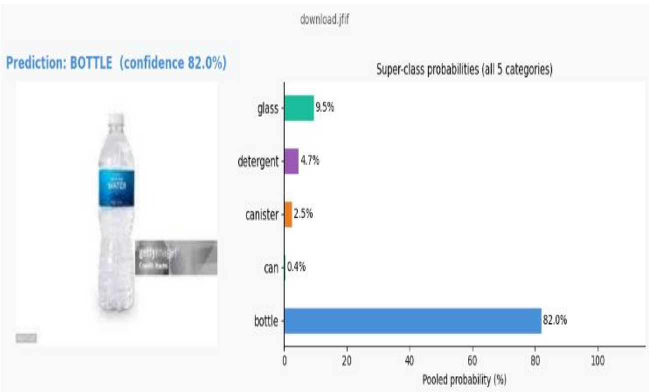

Figure 8: Waste material detector output. Uploaded image classified as BOTTLE (82.0%). Bar chart shows pooled super-class probabilities.

VI. Conclusion And Improvements

A. Conclusion

We successfully trained a ViT-B/16 classifier on the WaRP-C dataset, reaching 77.11% test accuracy, macro F1 of 0.7678, and a very strong AUC of 0.9808 across 28 fine-grained waste categories. The combination of Focal Loss, WeightedRandomSampler, LLRD, stochastic depth, and MixUp/CutMix augmentation was key to handling the class imbalance and achieving good generalization. The attention maps confirmed the model focuses on the waste object rather than background, which is a healthy sign.

The CLIP zero-shot experiment showed that off-the-shelf vision-language models are not suitable for this fine-grained task without fine-tuning - 0.52% accuracy versus 77.11% for our trained model. INT8 quantisation demonstrated that the model can be deployed on resource-constrained hardware with minimal accuracy loss, which is important for real industrial recycling systems.

Key limitations: the dataset is still relatively small (7,058 training images for 28 classes), and some rare sub-classes like bottle-oil-full (only 8 test images) are very hard to learn reliably. The Optuna search was limited to 3 short trials, so the hyperparameters may not be fully optimal.

G. *Future Work*

Future work could explore training on the full WaRP-D detection subset to extend the system beyond classification to localisation and WaRP-S (segmentation). Semi-supervised approaches leveraging unlabelled recycling plant footage could address the dataset size limitation. An ensemble of ViT-B/16 and ConvNeXt-Tiny may also improve performance given their complementary strengths - ViT leading on accuracy while ConvNeXt leads on recall. The WaRP dataset also includes a segmentation subset (WaRP-S) with pixel-level masks, which represents a natural extension for future work beyond classification.[1][2]

VII. Detection and WaRP-S Segmentation Advanced Extensions: WaRP-D

On top of the main WaRP-C classification task, we pushed things further by exploring detection and segmentation using the other two WaRP subsets. Object detection on WaRP-D tells the recycling system where each item is on the conveyor belt, while semantic segmentation on WaRP-S goes a step further and gives pixel-level boundaries. Together with classification, these three tasks form a complete automated waste recognition pipeline. For detection we used YOLOv8L and for segmentation we used DeepLabV3+ with an EfficientNet-B3 as backbone - both fine-tuned from ImageNet pre-trained weights, keeping the approach consistent with our main classification experiments.[4][5]

A. Discussion

Putting all three tasks together gives a clear picture of what a full automated recycling system looks like. Classification (WaRP-C) answers "what is this item?", detection (WaRP-D) answers "where is it?", and segmentation (WaRP-S) answers "exactly which pixels belong to it?". Our results show that the same main concepts - ImageNet pre-training, strong augmentation, and careful training design - transfers well across all three tasks. The efficiency numbers for YOLOv8L (26 img/sec at 83.6 MB) confirm it is practical for real conveyor deployment. For segmentation, the bigger challenge is data: 112 images are genuinely tight, and collecting more labelled segmentation data would likely push mIoU well above 0.70. An interesting direction for future work would be combining detection and segmentation into a single YOLOv8-seg model, which would simplify the overall system considerably.[5]

Table 5: YOLOv8L vs Prior Work on WaRP-D

Model	Precision	Recall	mAP50	mAP50-95
YOLOv8L (ours)	62.1%	54.5%	59.2%	46.2%

Detection and Segmentation experiments on WaRP-D and WaRP-S are detailed in the Appendix.

VIII. References

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IX. Appendix – Supplementary Diagrams and Result

Table 6: FP32 vs INT8 Efficiency Comparison

Metric	FP32	INT8	Change
Model Size (MB)	327.4	84.4	–74.2%
CPU Accuracy	85.00%	86.88%	+1.88 pp
Speed (ms/image)	356.83	304.82	1.2x faster

A. WaRP-D: Object Detection with YOLOV8L

Dataset

WaRP-D contains 2,452 training and 522 test images with YOLO-format bounding box annotations across the same 28 waste categories as WaRP-C. The training set has 8,823 total bounding box annotations. Bottle-transp is by far the most common class with 1,432 annotations, while several minority classes sit well below 150 - the same imbalance problem we dealt with in classification. Figure 9 shows a sample of training images with their ground-truth bounding boxes.

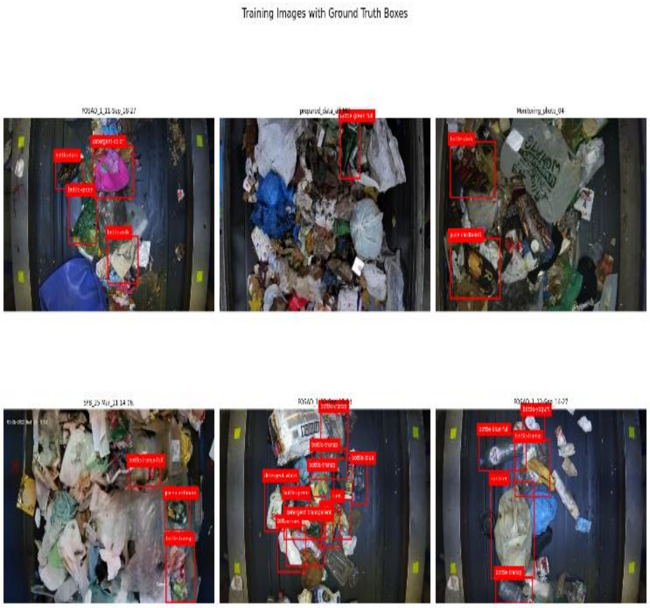

Figure 9: WaRP-D training images with ground-truth bounding boxes. Each box is colour-coded by class. The variation in lighting, scale, and occlusion is clearly visible.

Model and Training

We went with YOLOv8L - the large variant of the YOLOv8 family with 44M parameters - pre-trained on COCO. The choice of the large model over the small or medium variants was deliberate: we wanted to see how much performance a stronger backbone adds over the YOLOv8 baseline from Riyadi et al. [5]. Training ran for 20 epochs at 640x640 resolution with a batch size of 8 on a Tesla T4 GPU. We applied mosaic augmentation (combining 4 images per sample), MixUp, horizontal and vertical flips, HSV colour jitter, and 10-degree rotation to improve generalisation. Early stopping with patience 15 made sure we did not overtrain.

Results

Table I compares our YOLOv8L results against the previously published YOLOv5 and YOLOv8 baselines from Riyadi et al. [5]. Our model clearly outperforms both - mAP50 jumps to 59.2% compared to 54.6% for the prior YOLOv8 and just 40.6% for YOLOv5. Precision also improves significantly to 62.1%. The mAP50-95 of 46.2% shows there is still room to improve at tighter IoU thresholds, which is expected given the visual similarity between many wastes sub-classes. Figure 10 shows actual predictions on unseen test images.

Table 7: YOLOv8L vs Prior Work on WaRP-D

Model	Precision	Recall	mAP50	mAP50-95
YOLOv8L (ours)	62.1%	54.5%	59.2%	46.2%
YOLOv8 [5]	44.2%	51.3%	54.6%	-
YOLOv5 [5]	47.8%	56.9%	40.6%	-

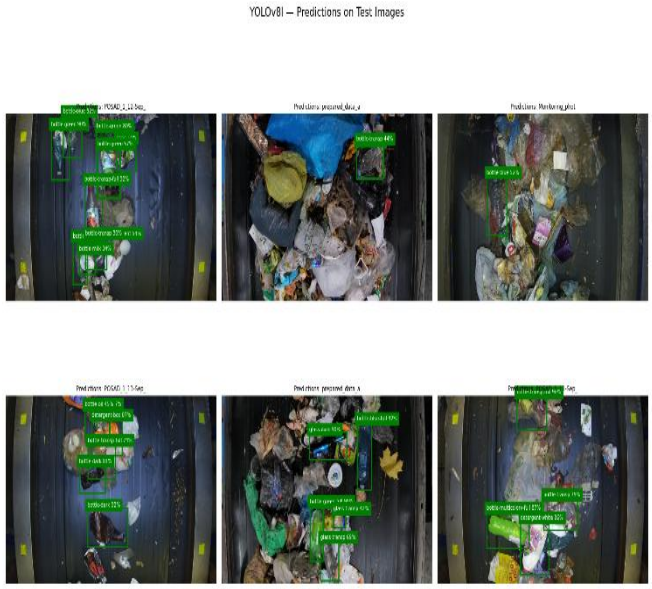

Figure 10: YOLOv8L predictions on unseen WaRP-D test images. Green bounding boxes show predicted regions with class name and confidence score. The model correctly localises most objects even under partial occlusion.

Efficiency

At 83.6 MB on disk and 37.9 ms per image, YOLOv8L achieves 26 images per second on a T4 GPU - comfortably above what a real-time conveyor sorting system would need. This means the model is ready for deployment without any further compression or quantisation.

B. WaRP-S: Semantic Segmentation with DeepLabV3+

Dataset

WaRP-S provides 112 images with pixel-level segmentation masks across 39 object classes. It is a small dataset by any measure, which made generalisation the main challenge throughout. Images were resized to 128x128 and the dataset was split 80/20 into train and val sets, giving 45 training batches and 11 validation batches at batch size 2.

Model and Training

We used DeepLabV3+ from the segmentation-models-pytorch library with an EfficientNet-B3 encoder, pre-trained on ImageNet. DeepLabV3+ uses atrous convolutions and an ASPP module to capture multi-scale context, which is important for segmenting waste objects that appear at different sizes and distances on the conveyor. The loss function was CrossEntropyLoss with ignore_index=255 to skip boundary pixels. We used AdamW with a learning rate of 1e-4 and cosine annealing over 30 epochs. Albumentations handled augmentation on both images and masks jointly, which torchvision cannot do.

Results

The model learned fast and steadily. mIoU jumped from 0.031 at epoch 1 to 0.671 at epoch 10 - the best checkpoint - and stayed stable after that, as shown in Table II and Figure 11. Train and val loss both decreased smoothly with no sign of overfitting, which is impressive given how small the dataset is. This really shows the value of ImageNet pre-training in low-data regimes.

Table 8: DeepLabV3+ Segmentation Results on WaRP-S

Model	Train Loss	Val Loss	Best mIoU	Epochs
DeepLabV3+ (ours)	0.4190	0.5920	0.6705	30

Best mIoU achieved at epoch 10 out of 30 total training epochs.

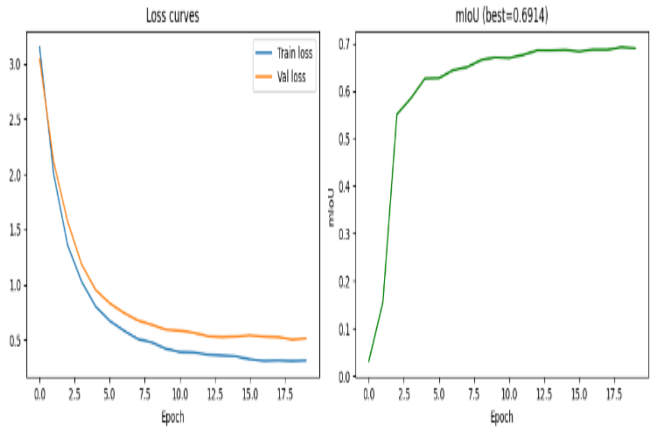

Figure 11: DeepLabV3+ training curves on WaRP-S. Left: train and val loss over 30 epochs. Right: mIoU over 30 epochs. Best mIoU of 0.671 reached at epoch 10.

Qualitative Analysis

Figure 12 shows predictions on four validation images alongside their ground-truth masks. The model does well on isolated objects with clear boundaries. Where it struggles is when two visually similar items are placed right next to each other - the predicted mask tends to merge them into one region rather than separating them cleanly. This is the main failure mode and is expected given the 112-image training set.

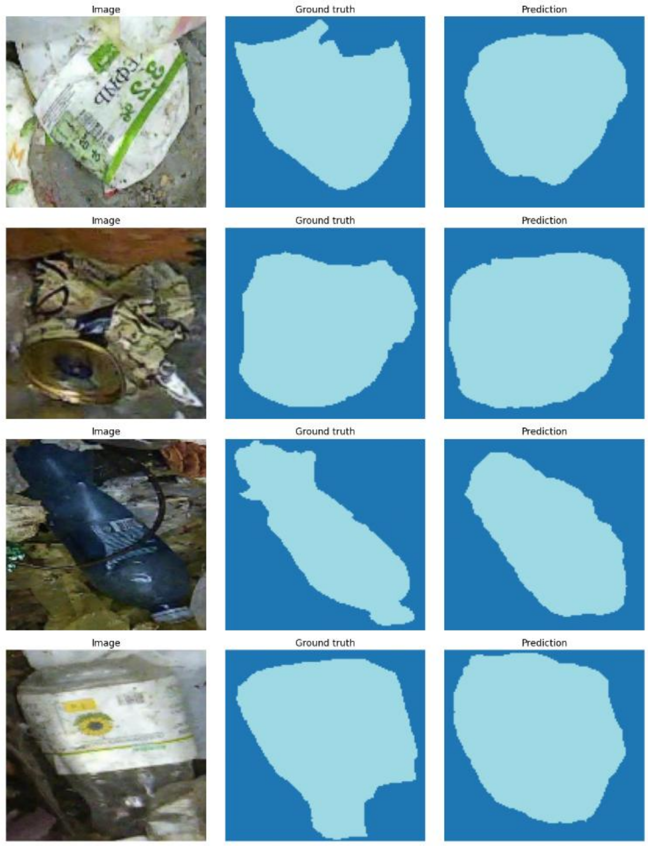

Figure 12: DeepLabV3+ predictions on WaRP-S validation images. Each row shows the input image (left), ground-truth mask (centre), and predicted mask (right). Colours correspond to object classes.

Table 9: Model Efficiency Summary

Metric	YOLOv8L	DeepLabV3+
Parameters	43.6M	~12M
Model size	83.6 MB	—
Inference speed	37.9 ms/image	—
Throughput	26 img/sec	—